

Complex video

Moving in the world frame: 16 persons, **the stroller**. Staying in place: 5 signs, 2 persons.

The stroller moves about 3.8 m in the world frame at an average speed of 1.04 m/s; it reverses direction around frame 4.

Question

Right answer: A

In the world frame (ignoring camera motion), what was the average speed (in meters per second) of **the stroller** between frame 5 and frame 16 of 34?

A. 0.70 B. 1.7 C. 0.95 D. 1.3

reference: 0.64 m/s

Qwen3-VL-8B · frames only

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<evidence>
object: the stroller
position_frame_5_m: (-0.7, 1.3, 8.3)
position_frame_16_m: (-0.6, 1.1, 6.2)
displacement_m: 2.1
time_s: 2.20
speed_mps: 0.95
</evidence>
<answer>C</answer>
```

Ours · frames + trajectory tokens

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<evidence>
object: the stroller
position_frame_5_m: (-0.6, 1.5, 8.7)
position_frame_16_m: (-0.6, 1.3, 6.9)
displacement_m: 1.8
time_s: 2.20
speed_mps: 0.80
</evidence>
<answer>A</answer>
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